
MATLAB/Simulink-based transient stability analysis of a multimachine power system

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Abstract Simulink is advanced software by MathWorks Inc., which is increasingly being used as a basic building block in many areas of research. As such, it also holds great potential in the area of power system simulation. In this paper, we have taken a multi-machine power system example to demonstrate the features and scope of a Simulink-based model for transient stability analysis. A self-sufficient model has been given with full details, which can work as a basic structure for an advanced and detailed study.

Keywords MATLAB; power system modelling; Simulink; transient stability

The stability of power systems has been and continues to be of major concern in system operation. Modern electrical power systems have grown to a large complexity due to increasing interconnections, installation of large generating units and extra-high voltage tie-lines etc. Transient stability is the ability of the power system to maintain synchronism when subjected to a severe transient disturbance, such as a fault on transmission facilities, sudden loss of generation, or loss of a large load. The system response to such disturbances involves large excursions of generator rotor angles, power flows, bus voltages, and other system variables. It is important that, while steady-state stability is a function only of operating conditions, transient stability is a function of both the operating conditions and the disturbance(s).¹ This complicates the analysis of transient stability considerably. Repeated analysis is required for different disturbances that are to be considered. In the transient stability studies, frequently considered disturbances are the short circuits of different types. Out of these, normally the three-phase short circuit at the generator bus is the most severe type, as it causes maximum acceleration of the connected machine.²

Historically, simulation of transient phenomena related to power systems has been carried on using the electromagnetic transients program (EMTP)³ or one of its variants, such as the alternative transient program (ATP) or electromagnetic transients for d.c. (EMTDC), which are all based on the trapezoidal integration rule and the nodal approach. SPICE is a general-purpose circuit simulation program, which was developed at the University of California, Berkeley.⁴ It contains models for basic circuit elements (R, L, C, independent and controlled sources, transformer, transmission line), switches and most common semiconductor devices: diodes, bipolar junction transistors (BJTs), junction field effect transistors (JFETs), MESFETs and MOSFETs. SPICE is mainly applied to simulate electronic and electrical circuits for different analyses, including d.c., a.c., transient, zero pole, distortion, sensitivity, and noise. SPICE uses the nodal approach with a variable-time-step integration algo-

rithm, so that it can correctly simulate switching power electronic circuits. The simulation of control systems in PSPICE A/D (a commercial version of SPICE by MicroSim) is facilitated by using the analog behavioral modeling (ABM) blocks. However, there are no specific models for power systems and drives, such as electrical machines, circuit breakers, surge arresters, thyristors, etc. To simulate a power system, the user has to build the needed models using SPICE primitives and basic elements, so the simulation setup can be highly time consuming.

Simulink is an interactive environment for modelling, analysing, and simulating a wide variety of dynamic systems. Simulink provides a graphical user interface for constructing block diagram models using 'drag and drop' operations.⁵ A system is configured in terms of block diagram representation from a library of standard components. A system in block diagram representation is built easily and the simulation results are displayed quickly. Simulation algorithms and parameters can be changed in the middle of a simulation with intuitive results, thus providing the user with a ready-access learning tool for simulating many of the operational problems found in the real world. Simulink is particularly useful for studying the effects of non-linearity on the behaviour of the system, and as such, is also an ideal research tool. The key features of Simulink are:⁶

- interactive simulations with live display;
- a comprehensive block library for creating linear, nonlinear, discrete or hybrid multi-input/output systems;
- seven integration methods for fixed-step, variable-step and stiff systems;
- unlimited hierarchical model structure;
- scalar and vector connections;
- mask facility for creating custom blocks and block libraries;

The user can also derive many features and in-built components from the Power System Blockset (PSB).⁷ PSB by itself gives the detailed three-phase representation of machine models and other components. Considering the overall complexity and data requirements (which might not be available in many cases) of a complete three-phase representation as required with PSB, we have considered its parent software package Simulink as a main tool in our present study. Excitation systems, turbine and governor blocks from PSB can be readily used with Simulink blocks as and when required. The user also has access to numerous design and analysis tools provided in MATLAB and its toolboxes.

Use of Simulink is rapidly growing in many areas of research work and so also in the field of power systems.⁸⁻¹⁰ In this paper we have demonstrated a simplified and yet an efficient approach to study the transient stability performance of a practical power system, with Simulink as a tool. We hope that this attempt will add some more practical information in this important and unexhausted domain.

Illustrative system example

We have considered the popular Western System Coordinated Council (WSCC) 3-machine, 9-bus system shown in Fig. 1.¹¹ This is also the system appearing in ref-

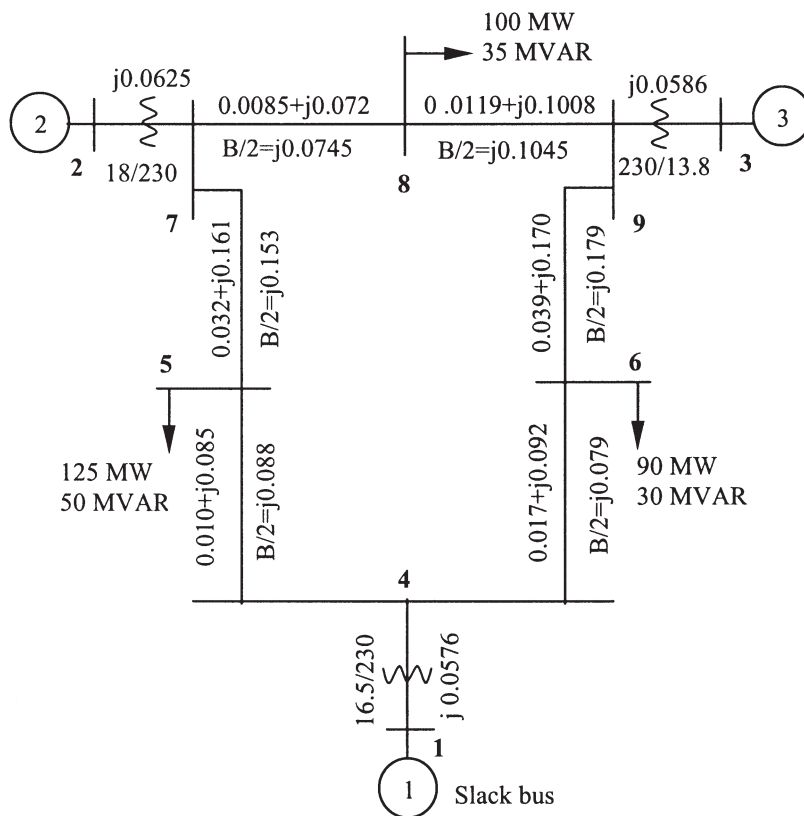


Fig. 1 WSCC 3-machine, 9-bus system; all impedances are in pu on a 100 MVA base.

erences [12] and [13] and widely used in the literature. The base MVA is 100, and system frequency is 60 Hz. The system data are given in Appendix I. The system has been simulated with a classical model for the generators. The disturbance initiating the transient is a three-phase fault occurring near bus 7 at the end of line 5–7. The fault is cleared by opening line 5–7. The system, while small, is large enough to be nontrivial and thus permits the illustration of a number of stability concepts and results.

System modelling

The complete system has been represented in terms of Simulink blocks in a single integral model. It is self-explanatory with the mathematical model given below. One of the most important features of a model in Simulink is its tremendous interactive capacity. It makes the display of a signal at any point readily available; all one has to do is to add a Scope block or, alternatively, an output port. Giving a feedback signal is also as easy as drawing a line. A parameter within any block can be con-

trolled from a MATLAB command line or through an m -file program. This is particularly useful for a transient stability study as the power system configurations differ before, during and after fault. Loading conditions and control measures can also be implemented accordingly.

Mathematical modelling

Once the $\mathbf{Y}$ matrix for each network condition (pre-fault, during and after fault) is calculated, we can eliminate all the nodes except for the internal generator nodes and obtain the $\mathbf{Y}$ matrix for the reduced network. The reduction can be achieved by matrix operation with the fact in mind that all the nodes have zero injection currents except for the internal generator nodes. In a power system with n generators, the nodal equation can be written as:

$$\begin{bmatrix} I_n \\ 0 \end{bmatrix} = \begin{bmatrix} Y_{nn} & Y_{nr} \\ Y_{rn} & Y_{rr} \end{bmatrix} \begin{bmatrix} V_n \\ V_r \end{bmatrix} \quad (1)$$

Where the subscript n is used to denote generator nodes and the subscript r is used for the remaining nodes.

Expanding eqn (1),

$$I_n = Y_{nn} V_n + Y_{nr} V_r, \quad 0 = Y_{rn} V_n + Y_{rr} V_r$$

From which we eliminate V_r to find

$$I_n = (Y_{nn} - Y_{nr} Y_{rr}^{-1} Y_{rn}) V_n \quad (2)$$

Thus the desired reduced matrix can be written as follows:

$$Y_R = (Y_{nn} - Y_{nr} Y_{rr}^{-1} Y_{rn}) \quad (3)$$

It has dimensions $(n \times n)$ where n is the number of generators. Note that the network reduction illustrated by eqns (1)–(3) is a convenient analytical technique that can be used only when the loads are treated as constant impedances. For the power system under study, the reduced matrices are calculated. Appendix II gives the resultant matrices before, during and after fault.

The power into the network at node i , which is the electrical power output of machine i , is given by¹²

$$P_{ei} = E_i^2 G_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^n E_i E_j Y_{ij} \cos(\theta_{ij} - \delta_i + \delta_j) \quad i = 1, 2, 3, \dots, n \quad (4)$$

Where,

$$\begin{aligned} \bar{Y}_{ij} &= Y_{ij} \angle \theta_{ij} = G_{ij} + jB_{ij} \\ &= \text{negative of the transfer admittance between nodes } i \text{ and } j \\ \bar{Y}_{ii} &= Y_{ii} \angle \theta_i = G_{ii} + jB_{ii} \\ &= \text{driving point admittance of node } i \end{aligned}$$

The equations of motion are then given by

$$\frac{2H_i}{\omega_R} \frac{d\omega_i}{dt} + D_i\omega_j = P_{mi} - \left[E_i^2 G_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^n E_i E_j Y_{ij} \cos(\theta_{ij} - \delta_i + \delta_j) \right] \quad (5)$$

$$\text{and } \frac{d\delta_i}{dt} = \omega_i - \omega_R \quad i = 1, 2, \dots, n \quad (6)$$

It should be noted that prior to the disturbance ($t = 0$) $P_{mi0} = P_{ei0}$;

Thereby,

$$P_{mi0} = E_i^2 G_{ii0} + \sum_{\substack{j=1 \\ j \neq i}}^n E_i E_j Y_{ij0} \cos(\theta_{ij0} - \delta_{i0} + \delta_{j0}) \quad (7)$$

The subscript 0 is used to indicate the pre-transient conditions.

As the network changes due to switching during the fault, the corresponding values will be used in above equations.

Simulink models

Classical system model

The complete 3-generator system, given in Fig. 1, has been simulated as a single integral model in Simulink. The mathematical model given above gives the transfer function of the different blocks. Fig. 2 shows the complete block diagram of a classical system representation for transient stability study. The subsystems 1, 2 and 3 in Fig. 2 are meant to calculate the value of electrical power outputs for different generators; for example Fig. 3 shows the computation of the power output of generator 1.

The model also facilitates the choice of simulation parameters, such as start and stop times, type of solver, step sizes, tolerance and output options etc. The model can be run either directly or from the MATLAB command line or from an *m*-file program. In the present study, the fault clearing time, the initial values of parameters as well as the changes in network due to fault, are controlled through an *m*-file program in MATLAB.

Modelling of power system components

The classical system model represented above can be supplemented with other power system components for a detailed study or for implementation of the stability improvement measures. References [1] and [2] give the simplified and generic models for many such components and transient stability improvement schemes. The block diagram models can be simulated within the Simulink environment almost in the same form. However, the representation of the transfer functions in the form of an integrator and gain with unity feedback is more convenient, when initial conditions have to be specified. Figs 4 and 5 give the Simulink models of a mechanical hydraulic control (MHC) governing system and that of a single reheat tandem-compound steam turbine, respectively. The typical parameter values are

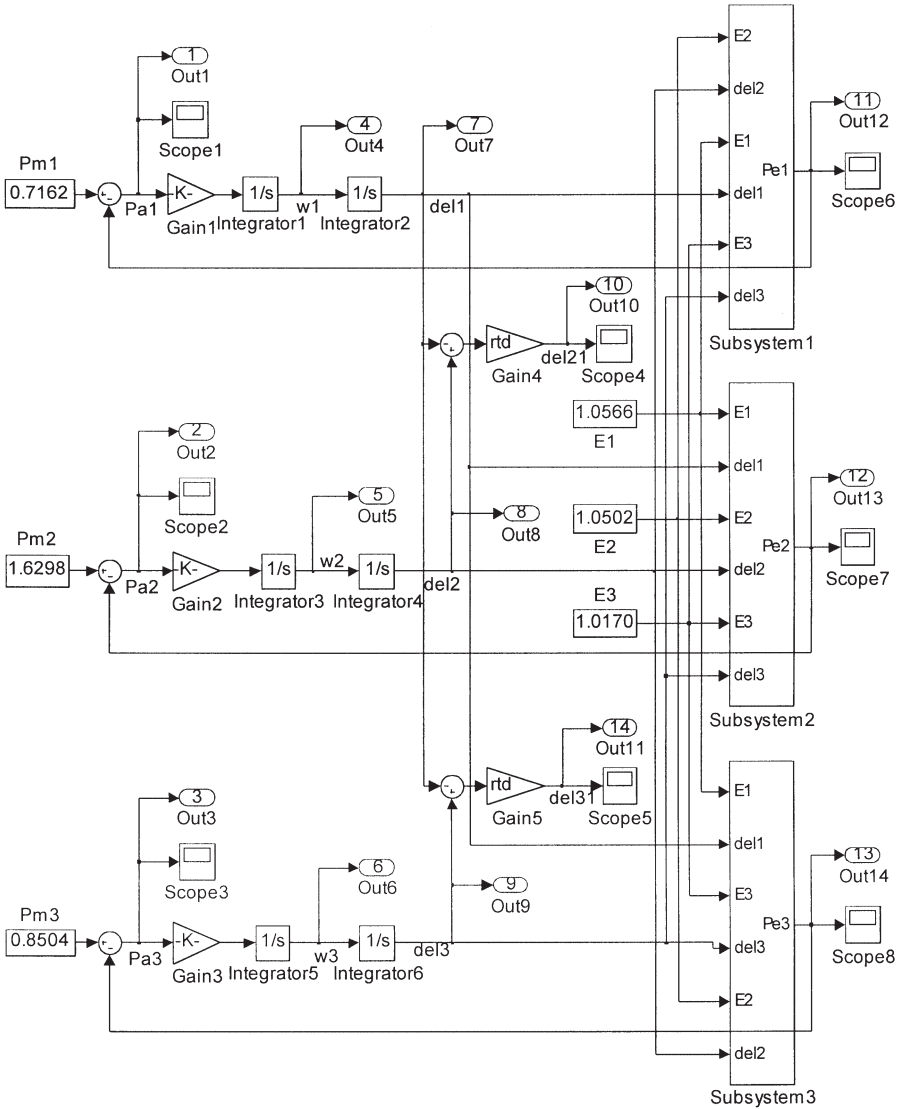


Fig. 2 Complete classical system model for transient stability study.

given in reference [1]. These values can be either defined in an *m*-file program or can be directly supplied to the Simulink models.

Simulation results

System responses are given for different values of fault clearing time (FCT). Figs 6(a) and (b) show the individual generator angles and the difference angles (with

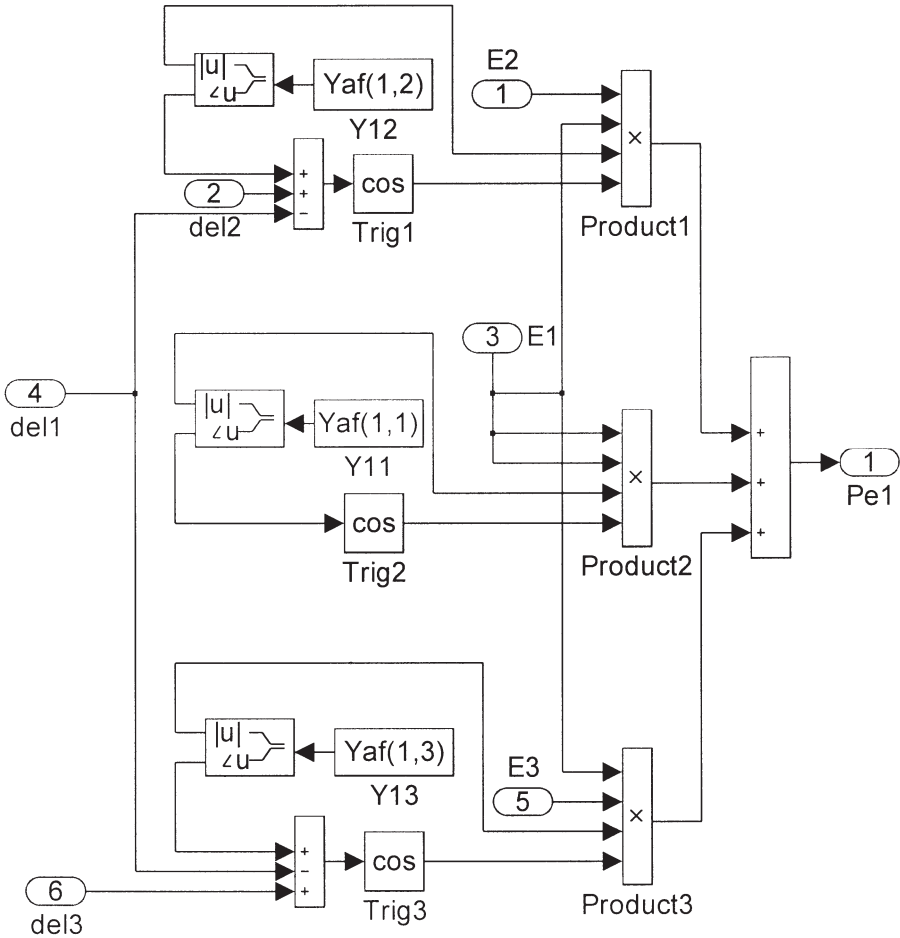


Fig. 3 Computation of electrical power output of gen. #1 by Subsystem 1.

gen. #1 as reference) for the system with FCT = 0.1 s, whereas Figs 6(c) and (d) show the rotor angular speed deviations and accelerating powers for the same case. The results show that the power system is stable in this case. We can see in the complete model of Fig. 2 that output ports 7, 8 and 9 give the individual generator angles of the respective machines. Ports 10 and 11 (or alternatively Scopes 4 and 5) give the relative angular positions of generators 2 and 3 respectively, with generator 1 as reference. Similarly, ports 4, 5 and 6 give the angular velocities of the machines, whereas Scopes 1–3 (or the corresponding ports) display the accelerating powers.

Figs 7(a), (b) and (c) show the system responses for a FCT value of 0.16 s. At this point the system is critically stable. The system becomes unstable for FCT = 0.17 s, as the system responses in Figs 8(a), (b) and (c) indicate.

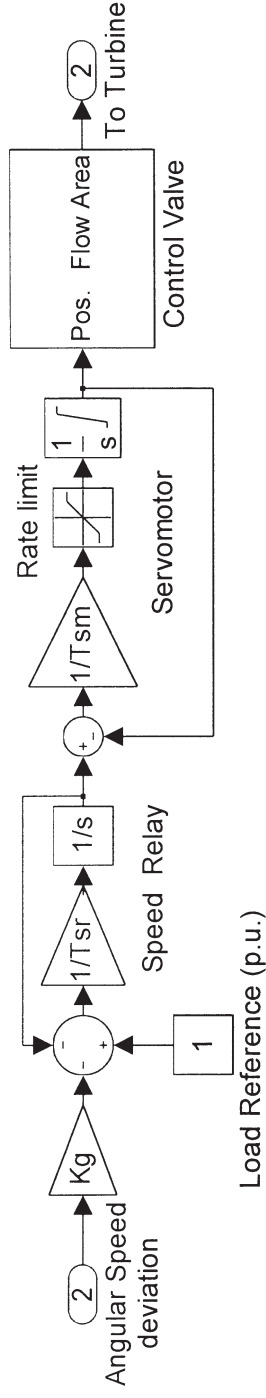


Fig. 4 Simulink model of MHC governor.

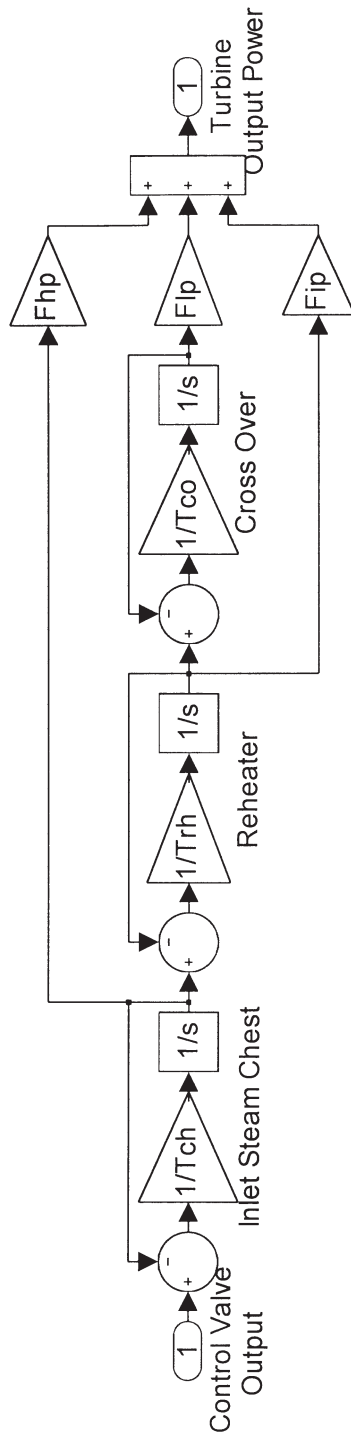
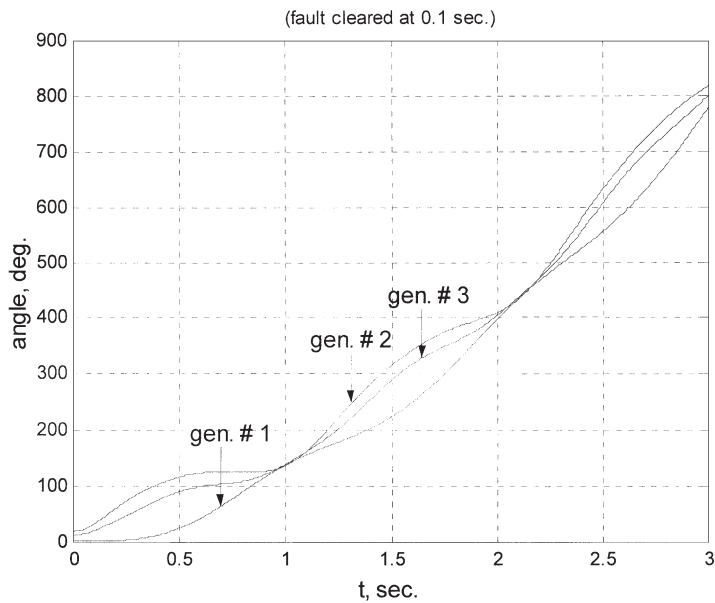
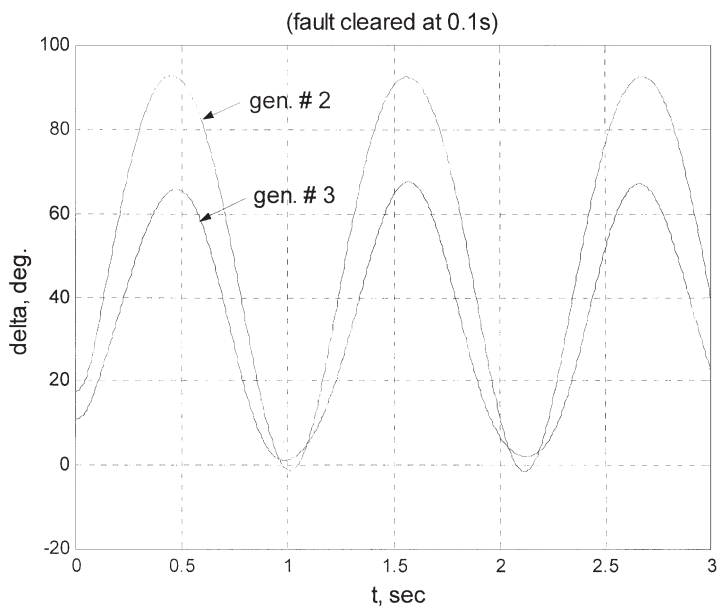


Fig. 5 Simulink model of single reheat tandem-compound steam turbine.

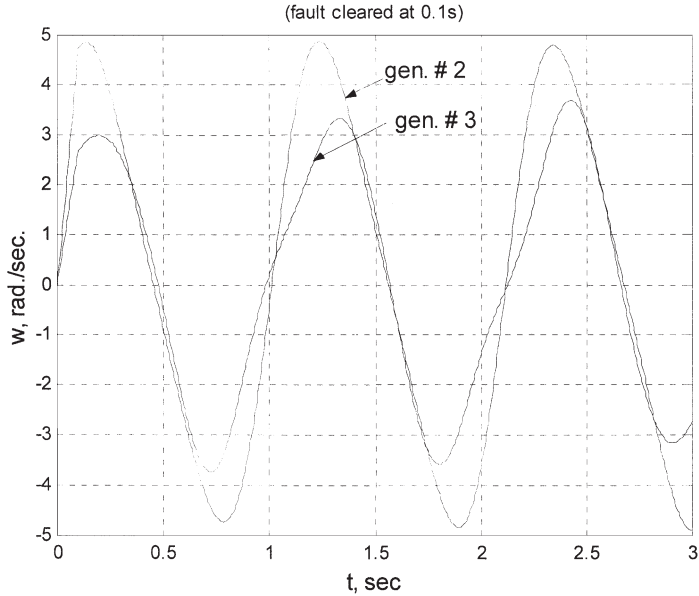


(a) Angular position of individual generators

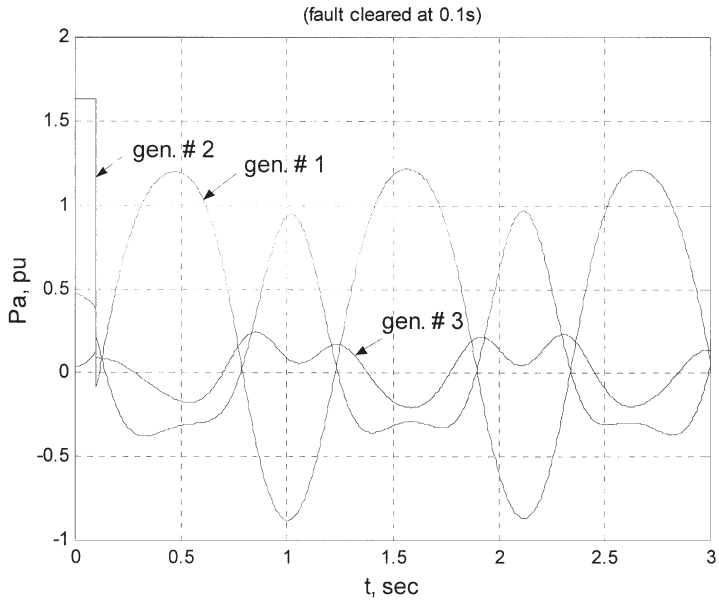


(b) Relative angular positions δ_{21} and δ_{31}

Fig. 6 (a–d) System responses for $FCT = 0.1$ s.

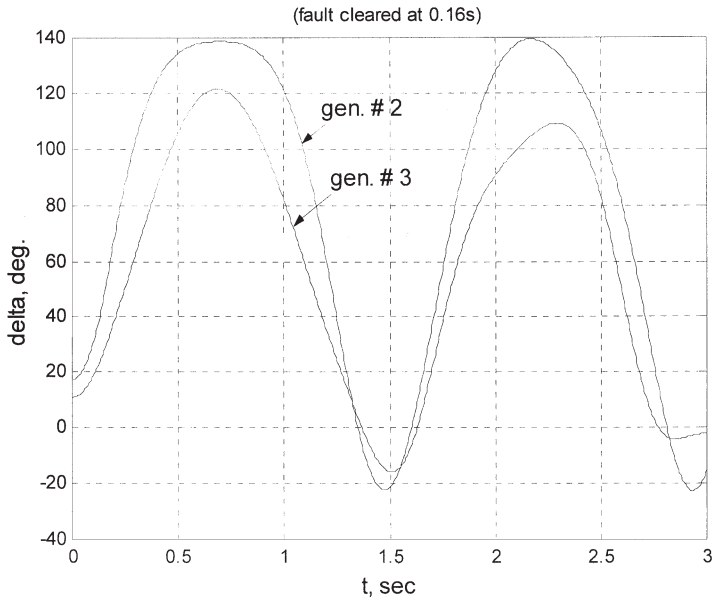


(c) Relative angular velocities ω_{21} and ω_{31}

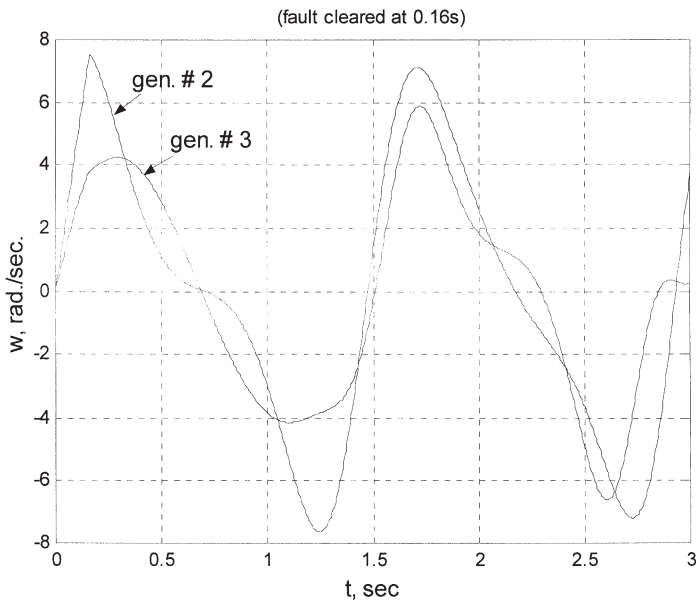


(d) Generator accelerating powers

Fig. 6 (continued)

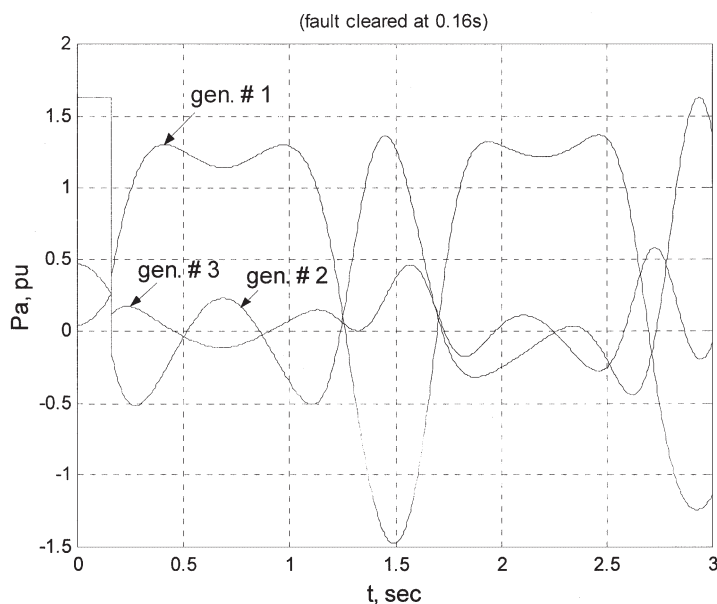


(a) Relative angular positions δ_{21} and δ_{31}



(b) Relative angular velocities ω_{21} and ω_{31}

Fig. 7 (a–c) System responses for $FCT = 0.16$ s.



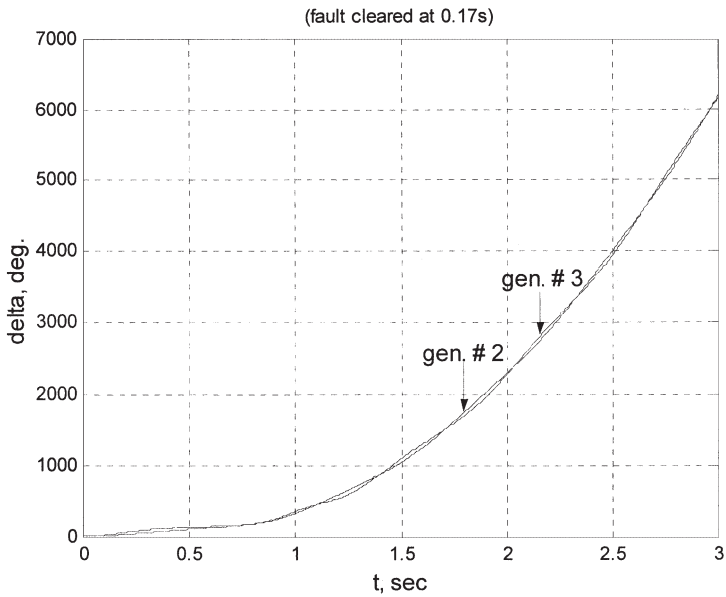
(c) Generator accelerating powers

Fig. 7 (continued)

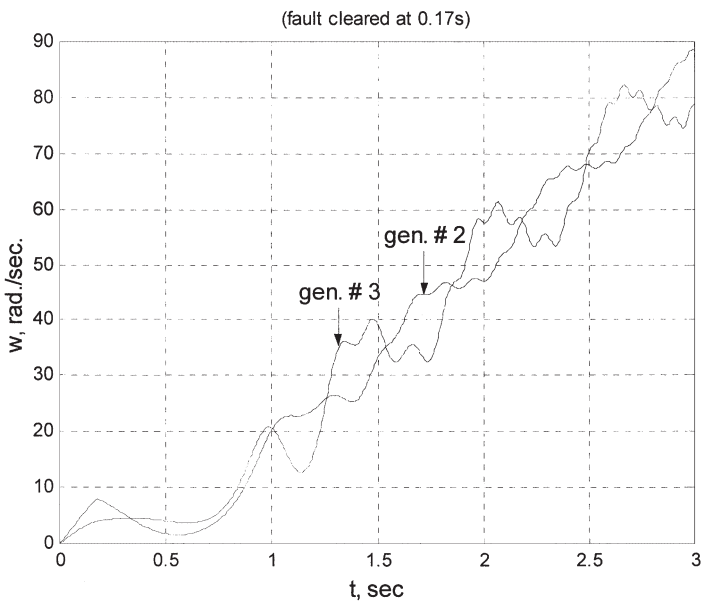
Thus a simple model based on Simulink is very well suited for analysing the transient stability performance of a power system under any system condition. The same model can also be extended to incorporate a more general (/practical) case of systems with exciters, turbines, speed governors etc.

Prospects of future work

It is clear from the above study that Simulink offers a wide perspective for simulation and analysis of various power system networks. The features of a Simulink model are exhaustive and at the same time it is very easy to understand and implement. In the present study, a simple classical model of a multi-machine system was considered. However, it explains very well the principles and the scope of the tool, typically for the study of transient stability in a power system. As indicated in the discussions in previous sections, the other factors such as effects of excitation, turbine, speed governor or any control measure, can be easily realised in a Simulink model, especially with the help of readily available and perfectly compatible tools like Power System Blockset. It should also be noted that a Simulink model can generate an equivalent C code for embedded applications and for rapid prototyping of control systems. Furthermore, the optimisation and application of advanced tools such as ANN and fuzzy logic, is also much easier as there are corresponding tool-boxes available within MATLAB.

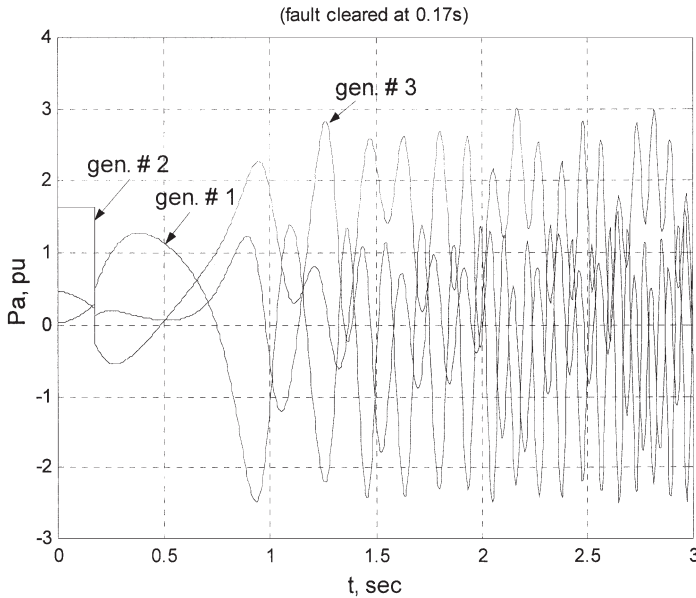


(a) Relative angular positions δ_{21} and δ_{31}



(b) Relative angular velocities ω_{21} and ω_{31}

Fig. 8 (a-c) System responses for $FCT = 0.17s$.



(c) Generator accelerating powers

Fig. 8 (continued)

Conclusions

A complete model for transient stability study of a multi-machine power system was developed using Simulink. It is basically a transfer function and block diagram representation of the system equations. A variety of component blocks are readily available in various Simulink libraries and also in other compatible toolboxes such as Power System Blockset, Controls Toolbox, Neural Networks Blockset etc. Thus a Simulink model is not only best suited for an analytical study of a typical power system network, but it can also incorporate the state-of-the-art tools for a detailed study and parameter optimization. A Simulink model is very user friendly, with tremendous interactive capacity and unlimited hierarchical model structure. Typically, for a transient stability study the model facilitates fast and precise solution of nonlinear differential equations viz. the swing equation. The user can easily select or modify the solver type, step sizes, tolerance, simulation period, output options etc. with the help of an appropriate menu from within Simulink. Any parameter within any block or subsystem of the model can be easily modified through simple MATLAB commands to suit the changes in the original power system network due to fault or a corrective action.

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Appendix I (generator data)

Generator no.	1	2	3
Rated MVA	247.5	192.0	128.0
kV	16.5	18.0	13.8
H (s)	23.64	6.4	3.01
Power factor	1.0	0.85	0.85
Type	Hydro	Steam	Steam
Speed	180 r/min	3600 r/min	3600 r/min
x_d	0.1460	0.8958	1.3125
x'_d	0.0608	0.1198	0.1813
x_q	0.0969	0.8645	1.2578
x'_q	0.0969	0.1969	0.25
x_l (leakage)	0.0336	0.0521	0.0742
T_{do}	8.96	6.00	5.89
T'_{go}	0	0.535	0.600
Stored energy at rated speed	2364 MW s	640 MW s	301 MW s

Note: Reactance values are in pu on a 100MVA base. All time constants are in seconds.

Appendix II (Reduced Y matrices)

Pre-fault network:

$$Y_{Rpf} = \begin{bmatrix} 0.8455 - 2.9883i & 0.2871 + 1.5129i & 0.2096 + 1.2256i \\ 0.2871 + 1.5129i & 0.4200 - 2.7239i & 0.2133 + 1.0879i \\ 0.2096 + 1.2256i & 0.2133 + 1.0879i & 0.2770 - 2.3681i \end{bmatrix}$$

During fault:

$$Y_{Rdf} = \begin{bmatrix} 0.6568 - 3.8160i & 0 & 0.0701 + 0.6306i \\ 0 & 0 - 5.4855i & 0 \\ 0.0701 + 0.6306i & 0 & 0.1740 - 2.7959i \end{bmatrix}$$

After fault network:

$$Y_{Raf} = \begin{bmatrix} 1.1386 - 2.2966i & 0.1290 + 0.7063i & 0.1824 + 1.0637i \\ 0.1290 + 0.7063i & 0.3745 - 2.0151i & 0.1921 + 1.2067i \\ 0.1824 + 1.0637i & 0.1921 + 1.2067i & 0.2691 - 2.3516i \end{bmatrix}$$